

2 Habituation is an example of learning that can be observed in many different animals.

- (a) Woodlice are small animals that live in damp places, for example under stones and in areas with decomposing leaves.

The photograph shows one species of woodlouse before and after being gently touched.



(Source: © StellaNature/Alamy Stock Photo)

Describe how you could investigate habituation in woodlice.

(4)

(b) Zebrafish are freshwater fish native to South Asia.

The photograph shows two zebrafish.



(Source: © Mirko Rosenau / Alamy Stock Photo)

If zebrafish are exposed to sudden sounds or flashes of light they show a 'startle reflex', where their body bends quickly and they swim away rapidly.

Scientists investigated habituation in zebrafish exposed to a sound.

Each fish was exposed to a sound and the distance travelled by the fish in 0.5 seconds was recorded.

Each fish was then exposed to the sound on four more occasions.

A control group of zebrafish was not exposed to any sound.

The table shows the results of this investigation.

Group	Mean distance travelled after stimulus / cm				
	1	2	3	4	5
Zebrafish exposed to sound	5.2	3.0	2.4	1.9	1.5
Zebrafish exposed to no sound (control group)	0.6	0.5	0.6	0.5	0.5

- (i) The mean distance travelled by the fish decreased from 5.2 cm to 1.5 cm after five stimuli.

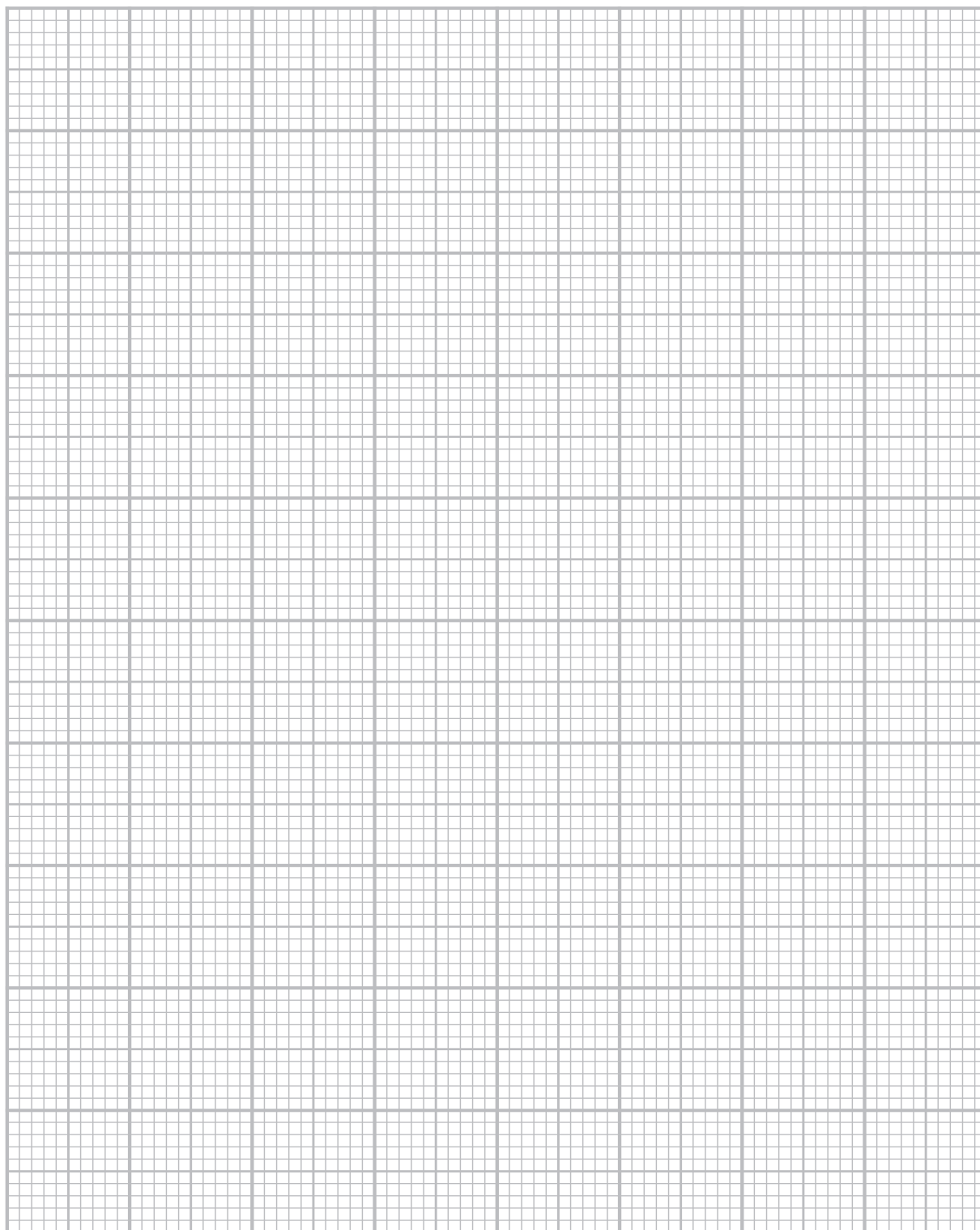
Calculate the percentage decrease in mean distance travelled.

(1)

Answer%

(ii) Plot a line graph for the results of this investigation.

(3)



(iii) Give **one** way that this investigation should be conducted to avoid harming the fish.

(1)

(iv) State **one abiotic** and **one biotic** variable that could affect this investigation of habituation in zebrafish.

(2)

Abiotic variable

Biotic variable

(v) Choose **one** of the variables you have identified in (iv).

Describe how this variable could be controlled and the effect it would have on the results if it is not controlled.

(2)

Variable

How this variable could be controlled.

The effect it would have on the results if it was not controlled.

(Total for Question 2 = 13 marks)